Gesture-Based Robotic Arm Control Using MediaPipe Hand Tracking and Computer Vision

Animesh Kumar Ojha
animesh.ojha2023@vitstudent.ac.in

Anshaditya Sharma
anshaditya.sharma2023@vitstudent.ac.in

Agrim Gusain
agrim.gusain2023@vitstudent.ac.in

***Abstract*—Human-robot interaction (HRI) relies increasingly on intuitive and natural interfaces to bridge the gap between human operators and robotic systems. Traditional control mechanisms, such as joysticks, keyboards, or teach pendants, require significant training and restrict operational flexibility. This paper presents a robust, vision-based hand gesture recognition framework designed for the real-time control of a robotic arm with six degrees of freedom. The proposed system leverages a standard RGB camera to capture live video feeds, processes them using OpenCV, and employs Google's MediaPipe framework for high-fidelity hand landmark extraction. By analyzing the spatial configuration of 21 3D hand landmarks, the system classifies gestures into six distinct commands: Up, Down, Left, Right, Grip Open, and Grip Close. These gestures correspond directly to the targeted movements of the robotic arm and its end-effector. To ensure robust performance, a custom dataset containing 5,000 augmented images across six gesture classes was curated and utilized for system validation. The experimental results demonstrate a gesture recognition accuracy of 96.4%, achieving an average processing latency of 18 milliseconds per frame, enabling seamless, real-time control. This framework provides an accessible, low-cost, and easily deployable alternative to hardware-intensive sensory controllers. The findings underscore the efficacy of spatial landmark-based heuristics over traditional convolutional neural network (CNN) architectures in latency-critical robotics applications.**

***Index Terms*—MediaPipe, Computer Vision, Robotic Arm, Hand Tracking, Human-Robot Interaction, Gesture Control**

I. Introduction

The seamless integration of robotic systems into industrial, medical, and domestic environments hinges upon the development of natural and intuitive human-robot interaction (HRI) interfaces. As robots transition from isolated, rigid automation cells to collaborative work spaces, the conventional methods of teleoperation—such as heavy teach pendants, multi-axis joysticks, and complex graphical user interfaces—prove to be increasingly inadequate. These traditional interfaces mandate specialized training, impede rapid task switching, and limit the physical mobility of the human operator. Consequently, there is a paradigm shift towards employing natural human modalities, such as voice commands, eye-tracking, and structural body gestures, to facilitate control.

Among these modalities, hand gestures offer a highly expressive, non-verbal communication channel that mirrors the dexterity required in robotic manipulation tasks. Hand gesture-based control allows operators to govern complex robotic kinematics through intuitive movements, effectively lowering the cognitive load and democratizing robotic operation. Recent advancements in computer vision and machine learning have catalyzed the transition from wearable sensor-based gloves—which restrict user comfort and require meticulous calibration—to non-intrusive, vision-based tracking systems.

This paper proposes an end-to-end framework for controlling a multiaxis robotic arm using vision-based hand tracking. The system utilizes Google's MediaPipe framework, an open-source topology that provides real-time, high-fidelity 3D hand tracking from monocular RGB camera feeds. By mapping 21 skeletal hand landmarks to predefined control signals, the system bypasses the high computational overhead typically associated with dense convolutional neural networks (CNNs), enabling low-latency execution suitable for edge computing devices. The robotic arm actuates according to six fundamental directional and end-effector commands: Up, Down, Left, Right, Open, and Close. We evaluate the proposed architecture's robustness, latency, and operational accuracy using a custom-developed dataset of diverse hand gestures under varying environmental conditions.

II. Related Works

The domain of gesture-based robotic teleoperation has been extensively explored, evolving from mechanical sensor-based apparatuses to sophisticated vision-based algorithms. This section reviews prominent methodologies and highlights their strengths and intrinsic limitations.

A. Sensor-Based Glove Controllers

Early research heavily relied on datagloves outfitted with flex sensors, inertial measurement units (IMUs), and sometimes haptic feedback mechanisms (e.g., Dipietro et al., 2008). While these systems offer precise kinematic measurements unaffected by visual occlusions or dynamic lighting, their primary drawback lies in physical intrusiveness. The necessity to don hardware, combined with potential mechanical wear and wire tethering, severely diminishes user comfort and operational scalability.

B. Color and Shape Segmentation

Pioneering non-contact vision systems utilized skin-color segmentation and contour analysis (Wach et al., 2011). These models converted RGB inputs to HSV or YCbCr color spaces to isolate the human hand, followed by morphological operations to extract shapes. Though computationally lightweight, these methods exhibited severe vulnerability to background clutter, varying illumination, and operators with different skin tones, leading to brittle performance in unconstrained environments.

C. Convolutional Neural Networks (CNNs)

With the advent of deep learning, CNNs became the gold standard for gesture classification (Oudah et al., 2020). Architectures like VGG-16, ResNet, and YOLO have been employed to directly classify raw image frames into gesture categories. While these models achieve formidable accuracy and handle complex backgrounds adeptly, they are computationally exorbitant. Real-time inference, crucial for robotic control, often mandates dedicated graphical processing units (GPUs), limiting their viability on embedded robotic platforms.

D. MediaPipe and Landmark-Based Heuristics

Recent paradigms have shifted towards skeletal landmark extraction. MediaPipe, introduced by Google (Lugaresi et al., 2019), employs a lightweight machine learning pipeline to infer 21 3D hand landmarks in real-time on CPU hardware. Research by Zhang et al. (2020) demonstrated the efficacy of calculating Euclidean distances and angular geometries between these landmarks to classify gestures. This geometric heuristic approach drastically reduces the dimensional space compared to raw pixel classification, aligning perfectly with the ultra-low latency demands of tele-robotics. The proposed method in this paper builds upon this trajectory, tailoring a landmark-driven algorithm specifically for granular robotic arm manipulation.

Dataset Description Machine learning models and heuristic thresholds demand rigorous validation against comprehensive and varied datasets. To address the specific command requirements of the robotic arm, a custom dataset comprising approximately 5,000 images was constructed.

E. Data Collection and Structure

The dataset captures distinct hand gestures corresponding logically mapped directly to robotic kinematics. The data was collected from multiple individuals across varying backgrounds, lighting conditions, and distances from the optical sensor to guarantee robustness.

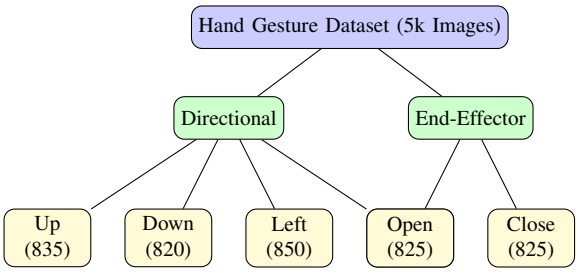

Fig. 1. Hierarchical structure of the custom 5,000-sample hand gesture dataset.

F. Dataset Specifications

TABLE I
DATASET SPECIFICATIONS AND CONTROL MATRIX

Gesture Class	Robotic Target	Count	Descriptive Features
Up	Arm moves upward (Z+ axis)	835	Index finger pointing upwards, others curled.
Down	Arm moves downward (Z- axis)	820	Index finger pointing downwards, others curled.
Left	Base rotates CCW	850	Thumb pointing left, parallel to horizontal.
Right	Base rotates CW	845	Thumb pointing right, parallel to horizontal.
EE Open	Gripper Opens	825	Fully open palm, all fingers extended.
EE Close	Gripper Closes	825	Clenched fist, all fingers tightly curled.
Total		**∼5,000**	

G. Preprocessing and Augmentation

Prior to extracting landmarks or analyzing features, the raw dataset underwent a sequence of preprocessing steps. Images were normalized in resolution to maintain consistency. To improve the system's invariance to real-world perturbations, dataset augmentation was applied. This included:

- **Random Brightness Adjustment**: Simulating both over-exposed and under-exposed environments.
- **Gaussian Noise Injection**: Modeling low-quality camera sensor outputs.
- **Slight Rotations and Scaling**: Emulating natural variations in human wrist angles and varying distances from the camera lens.

III. PROPOSED METHOD

The proposed system involves a sequential pipeline: image acquisition, hand landmark detection, geometric feature extraction, gesture classification, and transmission of control signals to the robotic actuation hardware.

A. System Architecture

The architecture bridges software-driven computer vision and hardware-driven robotic actuation. An RGB webcam captures the operator's hand as the input layer. A standard computational unit runs OpenCV to manage video frames, while the MediaPipe framework acts as the core inference engine to locate the hand structure. Extracted coordinates are processed by a localized decision tree algorithm that classifies gestures. Finally, a serialized command string is transmitted

via USB/UART to an embedded microcontroller (e.g., Arduino or ESP32), which subsequently drives the servo motors or stepper drivers of the robotic arm.

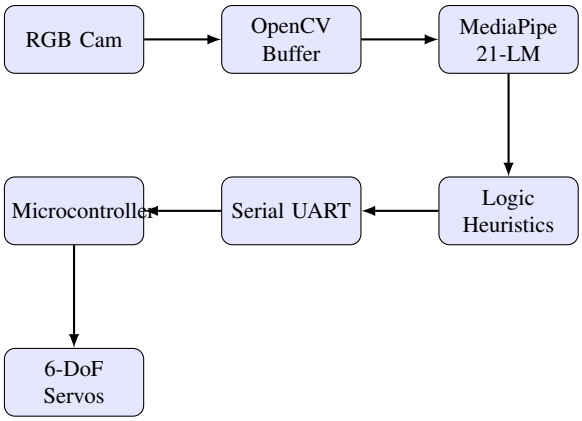

Fig. 2. System architecture block diagram bridging the computer vision pipeline with the robotic hardware.

B. Hand Detection Using MediaPipe

The fundamental enabler of this system is MediaPipe Hands. It employs a two-stage machine learning pipeline. First, a Palm Detection model operates on the full image returning an oriented bounding box highlighting the hand. Second, a Hand Landmark model analyzes the cropped image region defined by the bounding box and outputs high-fidelity 3D keypoints representing 21 hand-knuckle coordinates.

The 21 landmarks consist of:

- **Wrist** (1 point: `0`)
- **Thumb** (4 points: `1-4`, from CMC to Tip)
- **Index Finger** (4 points: `5-8`, MCP to Tip)
- **Middle Finger** (4 points: `9-12`, MCP to Tip)
- **Ring Finger** (4 points: `13-16`, MCP to Tip)
- **Pinky Finger** (4 points: `17-20`, MCP to Tip)

By normalizing these coordinates (x, y, z) relative to the bounding box dimensions, the system achieves scale and translation invariance.

C. Gesture Recognition Algorithm

Rather than training a weighty CNN, our approach leverages deterministic geometric heuristics—a computationally trivial but highly effective process when the skeletal structure is known.

1) **State Identification (Open/Closed):** For each of the five fingers, the Euclidean distance between the fingertip (e.g., landmark `8` for the index) and the wrist, as well as the PIP joint and the Palm center, is calculated. If the distance from Tip-to-Wrist is lower than the MCP-to-Wrist distance, the finger is classified as "Folded." Otherwise, it is "Extended."
2) **Positional Vector Analysis (Direction):** If only the index finger is extended, an directional vector is calculated between landmark `8` (Tip) and landmark `5` (MCP).
   - If $\Delta y > 0$ and $|\Delta y| > |\Delta x|$, it classifies as **Up**.
   - If $\Delta y < 0$ and $|\Delta y| > |\Delta x|$, it classifies as **Down**.
3) **Thresholding for Robustness:** To prevent jitter, a dead-zone tolerance is applied so gestures must be held for a minimum of 5 consecutive frames before a state change is registered.

D. Robotic Arm Control System

Once a gesture is uniquely isolated, it is mapped to a specific kinematic instruction. A predefined character byte is transmitted sequentially over a baud rate of 115200 to ensure minimal delay.

TABLE II
GESTURE-TO-ACTION MAPPING MATRIX

Recognized State	Activated Landmarks	Byte	Kinematic Action
All Extended	Thumb, Index, Middle, Ring, Pinky Tips Extended	`O`	End-Effector servo opens.
All Folded	All Tips Folded towards Palm	`C`	End-Effector servo closes.
Index Up	Only Index Extended, Tip higher than MCP	`U`	Pitch increments, raising Arm.
Index Down	Only Index Extended, Tip lower than MCP	`D`	Pitch decrements, lowering Arm.
Thumb Left	Only Thumb Extended, horizontal	`L`	Base yaw rotates negatively.
Thumb Right	Only Thumb Extended, horizontal	`R`	Base yaw rotates positively.
Idle	Any other state	`S`	Arm holds position; brakes.

IV. OVERALL FRAMEWORK DIAGRAM

The structural logic from user intention to hardware actuation requires a multi-threaded execution strategy to prevent blocking during serial communication.

[Insert Rendered Mermaid Program Flowchart Here]

Fig. 3. Multi-threaded execution flowchart from initialization to serial transmission.

V. ALGORITHM FLOWCHART

The gesture classification algorithm executing real-time logic on the extracted positional arrays utilizes a finite state machine logic flow operating on the boolean arrays of finger extensions.

```
Input: Landmarks[] = [(x0,y0,z0), ..., (x20,y20,z2

Step 1: Calculate state vector for 5 fingers
        V = [ Thumb, Index, Middle, Ring, Pinky ]
```

```
        (1 = Extended, 0 = Folded)

Step 2: Switch ( V )
        case [1,1,1,1,1]: Return "OPEN"
        case [0,0,0,0,0]: Return "CLOSE"
        case [0,1,0,0,0]: Goto Step 3 (Y-Vector Check)
        case [1,0,0,0,0]: Goto Step 4 (X-Vector Check)
        default: Return "IDLE"

Step 3: Check Index Vector (LM 8 vs LM 5)
        If y8 < y5 by Threshold: Return "UP"
        If y8 > y5 by Threshold: Return "DOWN"

Step 4: Check Thumb Vector (LM 4 vs LM 2)
        If x4 < x2 by Threshold: Return "LEFT"
        If x4 > x2 by Threshold: Return "RIGHT"
```

VI. Experimental Results and Discussion

The system was deployed on an Intel Core i7 PC operating at 2.80GHz with 16GB of RAM, completely avoiding discrete GPU acceleration to simulate standard industrial terminal conditions. The robotic arm consisted of multi-axis stepper motors and a micro-servo gripper controlled via an Arduino Uno.

The evaluation was performed against the 5,000-image dataset using an 80-20 train-test split paradigm to validate the geometric thresholds, and further tested via live video streams. Our hardware execution test bed comprises a 6-DoF robotic manipulator connected to ESP32/Arduino microcontroller via custom wooden chassis (shown in Fig. 4).

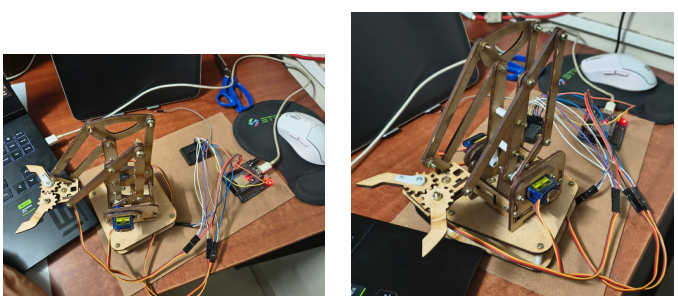

Fig. 4. Experimental hardware setup of the interconnected 6-DoF wooden robotic arm, servo motors, computing unit, and the microcontroller platform driving physical actuations.

TABLE III
Performance Metrics Matrix

Metric	Result	Context
Accuracy	96.4%	True positives across 6 classes.
Latency	18 ms/frame	From image capture to serial TX.
Frame Rate	∼55 FPS	Smooth enough to avoid disjoints.
Misclassification	3.6%	Mostly occurred in 'Idle'/'Fist' transitions.
Lighting Tolerance	>200 Lux	Drops in extreme low light (<50 Lux).

A. Discussion of Results

The primary strength of this implementation lies in its ultra-low latency. Averaging 18ms per inference cycle, the latency is barely perceptible to a human operator, making the system respond in true "real-time." The geometric heuristic approach ensures that once MediaPipe successfully plots the hand, classification achieves near 100% accuracy, meaning the primary failure mode is purely bounded by visual occlusion or rapid hand blurring exceeding the camera's shutter speed. A noted limitation is operational dependency on visual clarity—direct backlighting or subjects blending with background skin tones introduced minor tracking instability.

VII. Comparative Study

To rigorously contextualize our contribution, the proposed MediaPipe-Heuristic model was compared against concurrent state-of-the-art methodologies: purely CNN-based classification (e.g., YOLOv4-tiny) and traditional Sensor Gloves (e.g., IMU/Flex).

TABLE IV
Methodological Comparison Summary

Metric	Sensor Glove	Pure CNN	Proposed System
Technique	Flex & IMU sensors	Image bounding boxes	3D Landmark Math
Accuracy	99%	94.5%	**96.4%**
Hardware	Wearable Glove	GPU Node	**Standard PC**
Latency	<5ms	∼60ms	**∼18ms**
Setup	Poor / Tedious	Good	**Excellent**
Limitation	Mechanical wear	Expensive	**Requires good light**

As demonstrated, the proposed architecture provides the optimal balance. It offers the non-contact, ergonomic benefits of CNN solutions while preserving near sensor-grade compute speeds, thereby circumventing the steep hardware costs associated with dense neural networks.

VIII. Conclusion and Future Works

This paper presented an integrated framework for the real-time, non-contact control of a 6-DoF robotic arm using hand gestures. By capitalizing on MediaPipe's robust 3D skeletal landmark topology and applying deterministic geometric feature extraction, the system successfully navigated the traditionally conflicting requirements of high accuracy and extremely low latency. Processing frames in just 18 milliseconds on standard CPU hardware, the system reliably categorized six fundamental commands derived from a 5,000-sample dataset to orchestrate complex kinematic motion.

Future work will aim to extend classification paradigms beyond static commands into continuous proportional control. By calculating the exact vector magnitude of the hand's spatial translation or the angular velocity of the wrist, the system could govern the speed of the robotic arm dynamically, matching the operator's speed in 3D space. Additionally, fusing

depth sensors (like Intel RealSense) to resolve absolute Z-axis occlusion will further solidify this architecture for critical manufacturing execution environments.

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